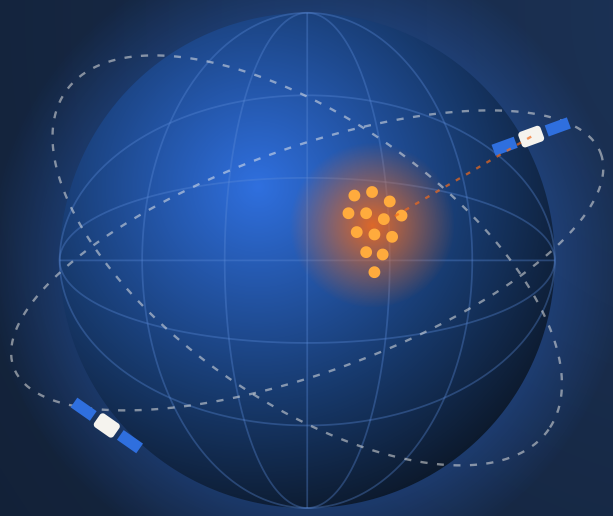
PROJECT GUIDE · OCTOBER 2026

Bhu-Drishti

Satellite monitoring for every district in India: ask in plain words, get an answer with its uncertainty stated, and send a report officials can act on.



THE TEAM

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Backend & APIs

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Satellite Data & AI Engine

OVERVIEW

Bhu-Drishti at a glance

Bhu-Drishti (“Earth vision”) turns free satellite data into answers district officials can use. A user types a question such as *“What percentage of cropland in Barpeta was flooded during 1–15 July 2024?”*. The system reads live Sentinel and NASA data and replies with a number, a confidence range, a map, the evidence it used and practical next steps. It can also watch an area on a schedule, raise alerts and email reports after a person approves them.

735
Indian districts with live data (geoBoundaries)

4
workflows: floods, fires, burned cropland, construction

6
live data providers behind one data layer

83
automated tests (65 engine + 18 backend)

Talk plan: about 18 minutes plus questions

	Section	What to get across	Time
1	The problem	Why officials can’t use satellite data today	2 min
2	What Bhu-Drishti does	The five-step journey from question to report	2 min
3	Workflows	Floods, fires, burned cropland, new construction	2 min
4	Data sources	Six providers, one secure data layer	1.5 min
5	Architecture	Web app → API → AEQE engine → data	1.5 min
6	**The AEQE engine**	Sampling, honest ranges, stopping rules, results	**5 min**
7	Product and reports	Pages, two PDFs, automatic email, approval	1.5 min
8	Security and quality	What makes it safe to run for real	1 min
9	Limits, roadmap, team	What is honest to claim and what comes next	1.5 min

> **Opening line:** “Every flood season, district officials have to decide where to send relief and how much crop loss to report, often before a field survey is possible. Bhu-Drishti lets them ask a satellite in plain language and tells them how sure the answer is.”

SECTION 1 2 MIN

The problem: data exists, answers don’t

Satellites photograph every Indian district every few days, and most of that data is free. Yet a district officer still can’t easily ask “how much farmland is under water?” and get a trustworthy number.

Disasters move fast

Floods and stubble fires change the picture in days. Field surveys arrive after the decisions about relief, compensation and enforcement have been made.

Data is scattered

Radar, optical imagery, fire hotspots and rainfall sit with different agencies (ESA, NASA, Microsoft, Google), each with its own accounts, formats and limits.

Images aren’t answers

Turning raw imagery into “how much, where, how sure” needs remote-sensing experts, which most district offices do not have.

Why a simple “run the model on everything” is not enough

- **Too slow:** a large district has thousands of tiles. Processing every one at full resolution takes far longer than an official can wait.
- **Costly:** provider quotas and processing units run out.
- **Overconfident:** a single number with no error range invites wrong decisions.
- **Hard to audit:** without a record of which scenes were used, nobody can check the answer later.

Our goal

Give any official, in any of India’s 735 districts, a **fast**, **honest** (with a stated range) and **traceable** answer about floods, fires and land change, then deliver it as a report they can forward up the chain.

> **Say:** “The bottleneck isn’t satellites. It’s turning pixels into a number someone can sign off on, quickly, and saying how sure we are.”

SECTION 2 2 MIN

What Bhu-Drishti does: from question to report

01 Ask
Type a question in your own words and pick a district or draw an area.

02 Understand
The compiler turns it into a precise plan: task, target, dates, precision.

03 Check
Validation confirms the data exists and the question is in scope.

04 Answer
AEQE samples satellite tiles and streams a tightening estimate live.

05 Report
Summary and technical PDFs, emailed after approval.

Questions it understands

People don't have to use set phrases. These all work:

- "Is farmland under water right now?"
- "Parali burning this week in Sangrur"
- "New houses since 2021"
- "km² of cropland flooded in August"
- "Where are the fires?"

Out-of-scope questions

Questions about air quality, weather, forests or cricket get a clear "not available here" reply with suggested questions, instead of an error.

If the question names another district, the app offers to **switch the area**.

What every answer contains

The number

For example *"about X % of cropland"*, with a **confidence range**, the area in km² and a per-tile map.

The evidence

Which satellite scenes were used, rainfall and MODIS flood context, and forecasts, kept separate from the number.

What to do

Recovery and prevention advice by hazard, season and state (crop insurance, SDRF norms, contingency plans).

> **Demo tip:** ask a flood question for a demo district. The interval narrows live on screen and stops by itself once it is tight enough; that is AEQE at work (Section 6).

SECTION 3 2 MIN

Four workflows, all across India

Workflow	What you get	How it is measured	Data
Flooded cropland	% of cropland ± percentage points, km², per-tile flood map	Radar change detection: water now that was not water in the dry season, on cropland, on gentle slopes (≤ 5°)	Sentinel-1 radar, ESA WorldCover, Copernicus DEM
Farm fires / stubble burning	Exact count of fire detections, ranked hotspot areas	Every detection counted; no sampling and no estimate needed	NASA FIRMS (VIIRS + MODIS)
Burned cropland (experimental)	km² burned ± %, per-tile burn share	Drop in a burn index (dNBR) between before and after images, cross-checked with fire detections	Sentinel-2 optical, NASA FIRMS
New construction	km² of new built-up area between two years	Dry-season images compared year to year with a built-up rule (NDBI / NDVI)	Sentinel-2 optical

Why radar for floods

Monsoon floods come with clouds. Sentinel-1 radar sees through clouds, and calm water looks dark to it. One threshold is chosen per region (Otsu's method, limited to −24…−14 dB) so every tile is judged the same way.

Crop recovery advice

When cropland is flooded or burned, the answer includes what farmers and officials can do next: report losses for insurance, re-sow short-duration varieties, and use soil and residue practices, matched to the hazard, season and state.

Honest about each method

- Burned cropland is labelled **experimental**: harvested paddy can look like a burn scar, so its thresholds still need calibration against field data.
- The built-up rule is spectral, not a trained model; dry riverbeds can look built.
- Two clearly labelled **simulated demo districts** with known ground truth let us test the statistics against the true answer.

> **Say:** "We picked these four because they are what district disaster and agriculture offices are asked about most, and because open satellite data can measure each of them."

SECTION 4 1.5 MIN

Six live data sources, one data layer

Planetary Computer

Microsoft. **Primary source:** Sentinel-1 radar, Sentinel-2 optical, ESA WorldCover land cover, Copernicus elevation.

Copernicus Data Space

ESA. A second Sentinel-1 catalogue and server-side statistics for cross-checks.

Google Earth Engine

AlphaEarth land embeddings, used as extra signals to guide sampling (optional).

NASA FIRMS

Near-real-time and archive fire detections. A complete catalogue, so fires are counted exactly.

NASA Earthdata

GPM IMERG rainfall and the MODIS near-real-time flood map, used as supporting context.

Open-Meteo

Rain and river-discharge forecasts (GloFAS), clearly labelled as forecasts.

What the unified data layer guarantees

- **Credentials never reach the browser:** keys live only on the server and are removed from logs and errors.
- **Provenance:** every answer records which source and scenes it used.
- **Status ladder:** each provider shows as working, degraded, not configured or failing in the app.
- **Retries and caching** soften network hiccups; repeat questions are cheap.
- **No silent guessing:** if data is missing or cloudy, the answer says “not enough evidence”.
- **Context is never mixed into the number:** rainfall and forecasts support the answer but don’t change it.

District boundaries: all 735 districts come from geoBoundaries (ODbL licence). Users can also upload their own area as GeoJSON or KML.

> **Say:** “Everything we use is open or free for public-good use. The work is in making six very different services behave like one reliable source.”

SECTION 5 1.5 MIN

System architecture

Web app
React, Vite and TypeScript. Leaflet maps, a three.js globe, light and dark themes.

API
Node.js and Fastify. Accounts, projects, monitors, alerts, reports, webhooks, audit log.

AEQE engine
Python and FastAPI. Question compiler, sampling, estimators, live progress stream.

Data sources
Six providers reached through one adapter layer with retries and caching.

Storage

SQLite by default (users, projects, sessions, reports); Postgres/PostGIS optional. Finished engine runs are saved to disk so results survive a restart.

Delivery

Reports are emailed through SMTP or handed to an **n8n** workflow, using signed webhooks (HMAC) so deliveries can't be forged.

Deployment

Docker Compose starts everything with one command. Only the website is public; the API, engine and n8n stay private behind it.

How one question flows through the system

#	Where	What happens
1	Web app	User signs in, picks a district and types a question
2	API	Checks the login token, enforces per-user limits and forwards the question to the engine
3	Engine	Compiles the question, validates data availability, builds the tile grid and starts sampling
4	Engine → app	Progress streams back live (Server-Sent Events): the estimate and range update after every batch
5	API	Saves the result, creates the report, renders both PDFs and emails them

> **Say:** "Three services, each with one job. The engine does the science, the API does people and paperwork, and the web app makes it usable."

SECTION 6 · THE AEQE ENGINE (1/4) 5 MIN TOTAL

AEQE: accurate answers without reading every pixel

AEQE is an **anytime, error-bounded query engine**. Instead of analysing every tile of a district at full resolution, it **scores every tile cheaply, analyses a statistically designed sample** with the expensive method, and **stops when the confidence range is tight enough**. The ranges stay valid however often the run is checked.

The expensive "oracle"

The full-resolution analysis of one tile (for example, radar flood detection on 10 m pixels). It is accurate but slow, so we want to run it as few times as possible.

The cheap "proxy"

A quick score for *every* tile from coarse whole-district reads: a ~200 m radar water overview, land-cover fractions, fire density and AlphaEarth embeddings.

The pipeline

1 Compile — Question → a typed query (task, target, dates, precision, confidence, budget)

2 Tile — Cut the district into ~0.01° tiles (about 1 km), clipped to its boundary; at most 8,000

3 Score — Compute the cheap proxy for every tile

4 Sample — Group by proxy, pick tiles in batches, run the oracle

5 Estimate — Update the estimate and range after each batch

6 Stop — Precision met, budget reached, or the user stops

Five kinds of question (tasks)

Task	Example	Precision means
Area fraction	% of cropland flooded	± percentage points (default ±3 pp)
Total	km² of new built-up area	± % of the estimate
Count	number of fire detections	± % of the estimate (exact for fires)
Find all	locate every tile with the target	a guaranteed **recall** (share of targets found)
Inspect	"is there flooding near this point?"	exact answers on a small area

> **Analogy:** "It's like an exit poll. You don't interview every voter; you choose a well-designed sample, and the maths tells you how far off you could be. We do that with satellite tiles, and we choose the sample smartly."

Smart sampling and the estimators

Step 1: stratify by the proxy

Tiles are ranked by their proxy score and split into **six strata** (groups), with finer groups at the top so rare targets get their own. Rank boundaries: 0, 30, 55, 75, 88, 96, 100 %.

Step 2: Neyman allocation

Each batch (24 tiles by default) goes where the uncertainty is largest: strata with more spread get more samples. Every stratum keeps a minimum share, so none is ignored. Within a stratum, tiles are drawn at random without replacement.



Orange: sampled tiles in high-proxy strata. Blue: sampled tiles in low-proxy strata. Dark: never analysed at full resolution, covered by the estimate.

Step 3: two estimators, the better one is reported

Stratified expansion estimator

Classic survey statistics (Cochran; Olofsson et al. for satellite area estimation):

$$\hat{Y} = \Sigma_h\, N_h \cdot \bar{y}_h$$
$$V = \Sigma_h\, N_h^2\, (1 - n_h/N_h)\, s_h^2 / n_h$$

N_h tiles in stratum h, n_h sampled, $\bar{y}_h$ and s_h^2 their mean and variance.

Cross-fitted difference estimator

A small ridge model predicts every tile's value from the cheap features; the oracle samples only **correct its bias** (prediction-powered inference, Zrnic & Candès 2024):

$$\hat{Y} = \Sigma_{i \in U}\, f(x_i) + \Sigma_h\, N_h \cdot \mathrm{mean}_{S_h}(y - f_{-k}(x))$$

Fitted on held-out folds, so the corrections are honest. Used once 30+ samples exist.

Key property: a good proxy makes the range much narrower for the same number of oracle runs, and a **bad proxy can only cost efficiency; it cannot bias the answer**, because the random oracle samples always correct it.

Knowing when to stop, honestly

AEQE checks its range after **every batch** and stops as soon as it is tight enough. But looking many times and stopping when the range looks good is exactly how normal statistics gets fooled. AEQE fixes this with **alpha spending**.

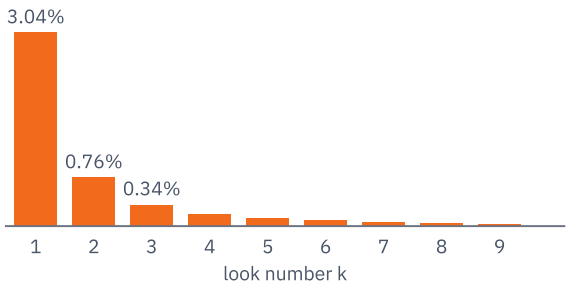
Alpha spending

The 5 % error budget (for 95 % confidence) is shared across all looks so it never exceeds 5 % in total:

$$\alpha_k = \alpha \cdot 6 / (\pi^2 \cdot k^2) \rightarrow \Sigma_k \alpha_k = \alpha$$

Look 1 gets the biggest share and later looks get less. Whenever the run stops (precision met, time or tile budget, or the user pressing stop), the reported range keeps its 95 % guarantee. The bound is conservative and needs no extra assumptions.

Error budget per look (α = 5 %)



Stopping rules

Precision met

The range is within the requested precision for **two batches in a row**, after at least 60 samples.

Budget reached

A time limit (default 120 s) or tile limit (default 1,000). It returns the best valid range so far.

User stop

“Anytime”: stop whenever you like and the range is still valid.

Special cases

Census for fires

FIRMS lists every detection, so AEQE counts all tiles exactly: no sample, no range.

Find-all with recall

Searches tiles in proxy order and audits the rest until an exact Clopper–Pearson bound certifies the recall target.

Says “I don’t know”

If tiles fail to read or are too cloudy, the result is **insufficient evidence**, never a made-up number.

Does it work? Benchmark results

Each row is **100 runs with different random seeds** on simulated districts, where we know the answer the oracle would give if it analysed every tile. “Covered” is how often the 95 % range contained that answer.

Question	Tiles	True answer	Mean estimate	Covered	Oracle tiles used	Fewer runs
Flood share ±3 pp (A)	1,719	37.5 %	37.5 %	97 %	157	**10.9×**
Flood share ±3 pp (B)	1,784	24.4 %	24.7 %	98 %	134	**13.3×**
Flood share ±1.5 pp (A)	1,721	14.3 %	14.3 %	98 %	257	6.7×
Flood share ±1.5 pp (B)	1,785	6.9 %	6.9 %	97 %	203	8.8×
New built-up ±10 % (A)	1,819	40.7 km²	40.2 km²	89 %	352	5.2×
New built-up ±10 % (B)	1,816	45.0 km²	45.0 km²	96 %	272	6.7×
Flood total ±10 % (A)	1,721	196.4 km²	197.3 km²	100 %	272	6.3×
Flood total ±10 % (B)	1,785	96.7 km²	97.2 km²	99 %	484	3.7×

3.7–13×
fewer expensive tile analyses than processing the whole district

97–100 %
coverage on flood questions at a nominal 95 %

Unbiased
mean estimates sit on the true answer

Reading the results honestly

- One row (built-up, district A) covered **89 %**, below 95 %. We report it rather than hide it; it points to a harder target for the proxy.
- The range covers **what the oracle would say over the whole district**. It does not cover the oracle’s own error against reality; that needs field data (see Limitations).
- Find-all kiln tests met their 90 % recall target in 100 % of runs. Kilns work only in the demo districts because no open kiln model exists for live data.

> **Say:** “In the flood case we analysed about 150 tiles instead of about 1,700, roughly ten times less work, and the range still contained the full answer 97–98 % of the time.”

SECTIONS 7 & 8 2.5 MIN

The product, and why it is safe to run

Dashboard
Alerts and recent activity at a glance

Explore
Ask a question and watch the live map and range

Areas & Monitors
Save areas and re-check them on every satellite pass

Alerts
Raised when a monitored value crosses a threshold

Reports
Draft, approve and track delivery of every report

Recipients
Officials who receive approved reports

History & Audit
Every question, delivery and change is logged

Data sources
Live status of all six providers

Summary PDF: for everyone
Headline in everyday words, how sure we are, a map of what was found, what it means and the top three actions.

Technical PDF: for experts
Method, confidence intervals, sampling details, data sources, scenes used and per-tile evidence.

Delivery: every new report is emailed automatically with both PDFs to a configured address. Officials receive it only after a person **approves** it, by SMTP or through an n8n automation, and every send is logged.

Security and quality

- Website public; the app **requires login**; passwords stored as bcrypt hashes
- Email password reset with single-use, 30-minute links
- Password change signs out other devices; “sign out everywhere”
- Rate limits on login and per-user limits on parallel analyses
- Security headers and a strict Content Security Policy
- Only the web container is exposed; health checks and auto-restart
- Daily backups with one-step restore
- CI runs 65 engine and 18 backend tests plus the build on every change

SECTION 9 1.5 MIN

Limitations, roadmap and likely questions

Honest limitations

- Ranges cover the method's full-district answer, not ground truth; **field validation** is still needed.
- Burned cropland is experimental until calibrated with reference data.
- Optical workflows depend on clear skies; radar floods do not.
- Brick kilns and point inspection work only in the demo districts.

Roadmap

- **Done (Phase 1):** live data, four workflows, monitors, alerts, two PDFs, email delivery.
- **Next (launch):** cloud server with HTTPS, professional email service, field-data calibration.
- **Phase 2:** WhatsApp and SMS alerts; roles for state, district and block teams.

Likely questions and short answers

Question	Answer
Why not process every tile?	Too slow and costly. Sampling gives the same answer within a stated range with roughly 4–13× fewer expensive analyses.
How do you know the range is right?	Repeated simulations with a known answer: the 95 % ranges covered it 97–100 % of the time on flood questions.
What if the cheap proxy is wrong?	Then the range is wider, not wrong. Random oracle samples correct the proxy's bias.
Why radar for floods?	Floods come with clouds; radar sees through them and calm water appears dark.
What does it cost to run?	The data sources are free or free for public-good use; it runs on one server with Docker.
Can anyone see my data?	No. Login is required, data is per user, keys never reach the browser and reports go out only after approval.

Thank you

github.com/aditimanthalkar53/Bhu-Drishti

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